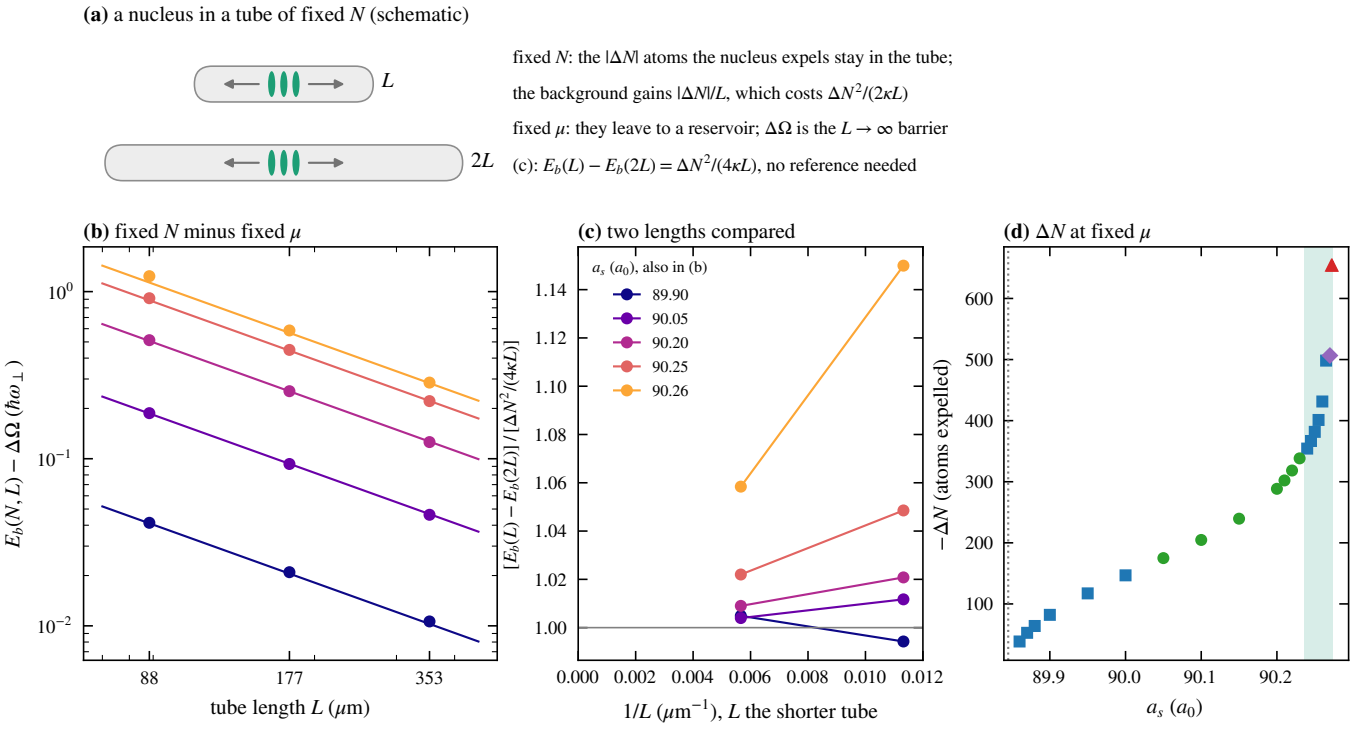
(a) a nucleus in a tube of fixed N (schematic)
L
2L
fixed N: the |ΔN| atoms the nucleus expels stay in the tube;
the background gains |ΔN|/L, which costs ΔN²/(2κL)
fixed μ: they leave to a reservoir; ΔΩ is the L → ∞ barrier
(c): E_b(L) − E_b(2L) = ΔN²/(4κL), no reference needed
(b) fixed N minus fixed μ
E_b(N, L) − ΔΩ (ħω⊥)
tube length L (μm)
(c) two lengths compared
a_s (a_0), also in (b)
89.90
90.05
90.20
90.25
90.26
[E_b(L) − E_b(2L)] / [ΔN²/(4κL)]
1/L (μm⁻¹), L the shorter tube
(d) ΔN at fixed μ
−ΔN (atoms expelled)
a_s (a_0)